TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

# CHAPTER 8: IONIC EQUILIBRIUM

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

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| **Board:** | Tamil Nadu State Board (TNSB) |
| **Syllabus:** | Samacheer Kalvi Textbook Curriculum aligned |
| **Standard:** | Class 12 English Medium |
| **Subject:** | Chemistry Volume 2 |
| **Chapter Name:** | Ionic Equilibrium |

**Learning Objectives:**

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

**Atomic Orbitals Wave Functions (s & p)**

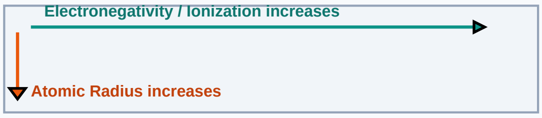
## Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

| Term | Official Definition & Context |
| --- | --- |
| **Lewis Acid and Base** | Acid is an electron pair acceptor (e.g. $BF_3$); base is an electron pair donor (e.g. $NH_3$). |
| **Buffer Solution** | A solution containing a weak acid/conjugate base that resists pH changes upon acid/base additions. |
| **Solubility Product** | The product of molar concentrations of constituent ions in a saturated salt solution, raised to powers. |

**Syllabus Exam Tip:** Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

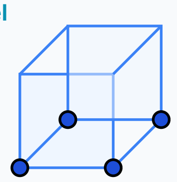
**Periodic Properties Trends Direction Map**



## Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

**Crystal Solid Lattice Unit Cell Model**



Simple Cubic Lattice System

### Essential Formulas, Laws & Identities:

- pH relation: pH = -log10[H+] | pH + pOH = 14
- Ostwald's Dilution: K_a = alpha^2 * C / (1 - alpha) = alpha^2 * C (for alpha << 1)
- Henderson-Hasselbalch: pH = pKa + log10([Salt] / [Acid])

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

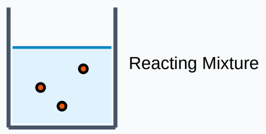
## Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

### Example 1: Calculate pH of a 10^-3 M HCl solution.

**Step-by-step Solution:**
HCl dissociates fully: [H+] = 10^-3 M.
pH = -log10(10^-3) = 3.

**Titration Beaker & Solute Precipitation**



## HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

### HOTS Challenge 1: Derive Henderson-Hasselbalch equation for an acidic buffer solution.

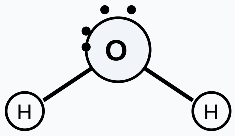
**Analytical Solution Strategy:**

Weak acid HA $\Leftrightarrow$ $H^+$ + $A^-$. $K_a = [H^+][A^-] / [HA] \Rightarrow [H^+] = K_a * [HA] / [A^-]$.

Take negative log: $-\log[H^+] = -\log(K_a) - \log([HA]/[A^-])$.

$pH = pKa + \log([A^-] / [HA]) = pKa + \log([Salt] / [Acid])$.

**Lewis Dot Structure ($H_2O$ Bent Molecular Geometry)**



## Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

**Q1 (1-Mark):** Explain conjugate acid-base pairs with an example.

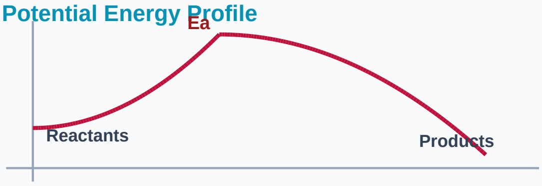
**Q2 (2-Mark):** State Ostwald's Dilution Law.

**Q3 (2-Mark):** Explain the common ion effect with a chemical example.

**Q4 (5-Mark):** What is buffer capacity? Write buffer index expression.

**Q5 (5-Mark):** Define solubility product Ksp and write expression for AgCl.

Chemical Reaction Potential Energy Profile



## Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

### Application Case: Biomedicine

Carbonic acid bicarbonate buffers maintain human blood pH at exactly 7.4.

### Application Case: Industrial Electroplating

Metal plating baths maintain strict pH limits to ensure uniform metal deposits.

**Cross-Curricular Link:** Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C6H6) Resonance

# Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

**Key Recall Tips:**

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, $cm^3$, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

**Syllabus Checklist:**

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

**CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!**

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

**Electrochemical Galvanic Cell System**